

Experiments on Sentence-level Confidence for GPT-3.5-Turbo and GPT-4-Turbo

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Here is the example of prompt for next experiment

```
{  
    'role': 'system',  
    'content': f"""You are a helpful assistant that corrects ASR errors. \  
You will be presented with an ASR transcription delimited by {delimiter} characters,\  
and your task is to rectify any errors in it. \  
Provide the most probable corrected transcription in string format. \  
Do not change the case, for example, lower case or upper case, in the transcription. \  
Do not output any additional text that is not the corrected transcription. \  
Do not write any explanatory text that is not the corrected transcription.  
"""  
}
```

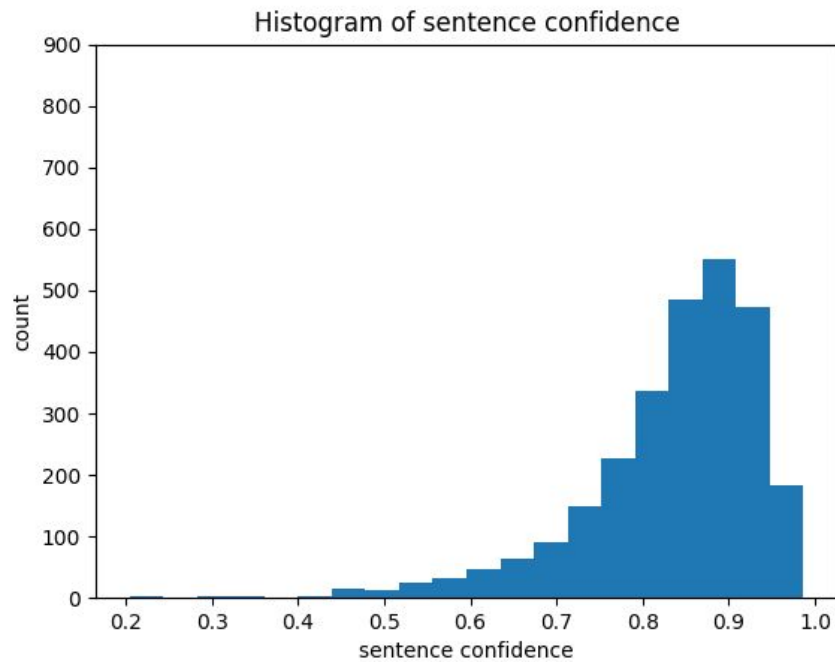
First step

We obtained the results, along with their associated sentence-level confidence scores, from the “exp_without_sentence_confidence” for both GPT-3.5-Turbo and GPT-4-Turbo.

Here is the one example of our output for this step

```
{  
  "asr_transcription": {  
    "text": " On the I'll do it more completely.",  
    "confidence_score": 0.79  
  },  
  "reference_transcription": "ONLY I'LL DO IT MORE COMPLETELY",  
  "corrected_asr_transcription": "I'll do it more completely."  
}
```

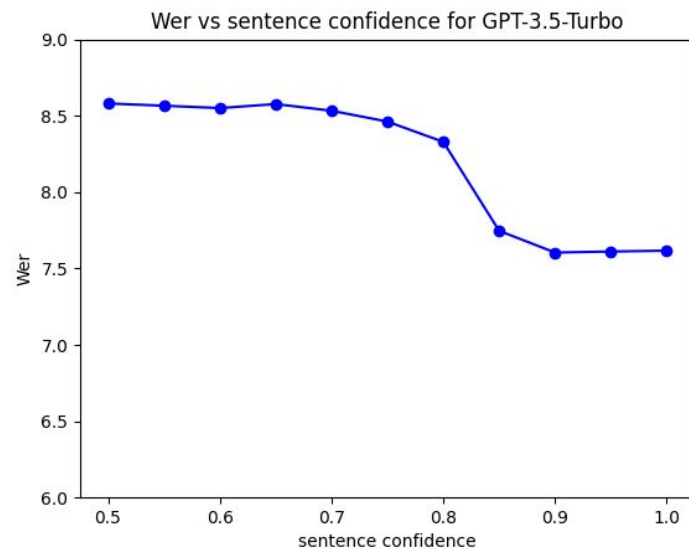
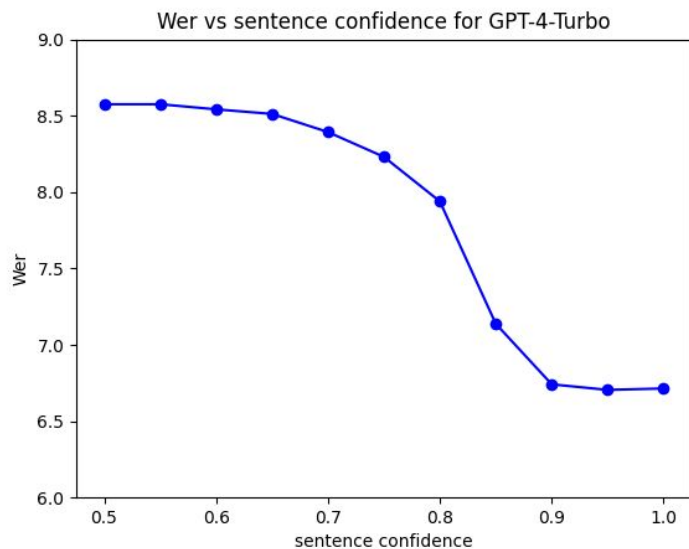
Histogram plot of the sentence-level confidence scores of the Librispeech transcriptions



Based on the histogram on last slide, we changed the range of threshold.

Threshold = [0.5, 0.55, 0.6, 0.65, ..., 1.0]

Wer vs sentence confidence plots for GPT-3.5-Turbo and GPT-4-Turbo



Thank you!